🎨 Project Design Phase: Streamlining Ticket Assignment for Efficient Support Operations

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1. Objective of the Design Phase

To translate project requirements and planned strategies into **detailed technical and process designs**, defining **how** the automated ticket assignment solution will function, integrate, and perform.
This phase ensures that the development team has a clear, validated blueprint before coding and implementation begin.

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2. Key Design Goals

- Design an **intelligent, automated ticket routing workflow**
- Ensure **seamless integration** with existing ticketing platforms (e.g., Zendesk, ServiceNow)
- Build a **scalable architecture** capable of handling increasing ticket volumes
- Maintain **data accuracy, privacy, and security**
- Create **user-friendly dashboards** for monitoring and control

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3. Design Activities

Activity	Description	Output
Requirements Validation	Review functional and non-functional requirements from planning phase	Confirmed Requirements List
Process Mapping	Create detailed ticket flow diagrams	Process Flowcharts
System Architecture Design	Define technical components and integration points	Architecture Diagram
Database & Data Model Design	Design data schema for tickets, agents, and skills	Entity-Relationship (ER) Diagram
UI/UX Wireframing	Design user interfaces for dashboards and control panels	Wireframes / Mockups
AI Model Design	Define NLP model for ticket classification and routing logic	ML Design Document

Activity	Description	Output
Security & Access Design	Define roles, permissions, and compliance rules	Security Specification
Prototype Definition	Plan and design a functional MVP for early testing	Prototype Blueprint

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4. Process Flow Design

Example: Automated Ticket Assignment Workflow

1. **Ticket Creation:**
   - Ticket submitted via email, chat, or form.
   - Basic data captured: issue type, customer, urgency, and keywords.
2. **Ticket Classification:**
   - NLP model analyzes ticket content and assigns category (e.g., Billing, Technical, Login).
3. **Priority Scoring:**
   - System assigns priority based on SLA, sentiment, and keyword indicators.
4. **Agent Matching:**
   - Skill-based matrix checks agent availability and expertise.
   - Workload balancing ensures even distribution.
5. **Assignment & Notification:**
   - Ticket auto-assigned to best-fit agent.
   - Agent and customer notified instantly.
6. **Monitoring & Escalation:**
   - Dashboard displays performance, SLA countdowns, and load per agent.
   - Escalations triggered automatically if thresholds exceeded.

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5. System Architecture Design

Proposed Architecture Layers:

1. **User Interface Layer**
    - Agent Dashboard
    - Admin/Manager Dashboard
    - Reports and Analytics Views
2. **Application Logic Layer**
    - Ticket Routing Engine
    - Skill & Workload Balancer
    - SLA Management Module
3. **AI/ML Layer**
    - NLP-based Ticket Categorization
    - Predictive Priority Model
    - Continuous Learning Loop (feedback from resolved tickets)
4. **Data Layer**
    - Ticket Database
    - Agent Skills Repository
    - Historical Ticket Logs
    - Configuration Tables
5. **Integration Layer**
    - Connects to existing ticketing systems (API-based)
    - Integrates with communication tools (Slack, Teams)

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6. Data Design

Entities:

- **Tickets:** ticket_id, category, priority, created_time, assigned_agent, SLA_time, status
- **Agents:** agent_id, name, skills, workload, availability
- **Skills Matrix:** skill_id, skill_name, proficiency_level
- **Routing Rules:** rule_id, condition, assigned_group

Relationships:

- One agent can handle multiple tickets
- Each ticket has one assigned agent
- Skills mapped to agents and categories

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7. User Interface Design

UI/UX Goals:

- Simplify agent experience with clear task visibility
- Provide real-time workload and performance indicators
- Offer intuitive admin controls for rule editing and monitoring

Main Interfaces:

- **Agent Dashboard:** Displays assigned tickets, priorities, and performance metrics.
- **Manager Dashboard:** Shows overall workload, SLA risks, and agent utilization.
- **Configuration Panel:** Allows modification of routing rules and skill mapping.

Deliverable: Wireframes or mockups created using tools like Figma or Adobe XD.

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8. AI/ML Model Design

Component	Description
Input Data	Historical tickets, categories, resolution notes, response times
Model Type	Natural Language Processing (Text Classification)
Algorithm	BERT / GPT-based classifier for ticket type prediction
Training Output	Probability scores for each category (e.g., Billing 85%)
Integration	Exposed via API to routing engine for real-time predictions
Feedback Loop	Model retrains based on ticket reassignment and feedback logs

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9. Security & Compliance Design

- **Authentication:** Single sign-on (SSO) or OAuth integration
- **Authorization:** Role-based access control (Agent, Manager, Admin)

- **Data Privacy:** PII masking and encrypted storage
- **Audit Trails:** Full log of ticket assignments and changes
- **Compliance:** GDPR / SOC 2 / ISO 27001 adherence

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10. Prototype Design Plan

Goal: Validate functionality of routing logic and AI categorization before full development.

Includes:

- Basic ticket input form
- NLP-powered classification
- Automated rule-based assignment
- Simplified dashboard for tracking assignments

Tools: Python Flask + PostgreSQL + Figma (for UI mockups)

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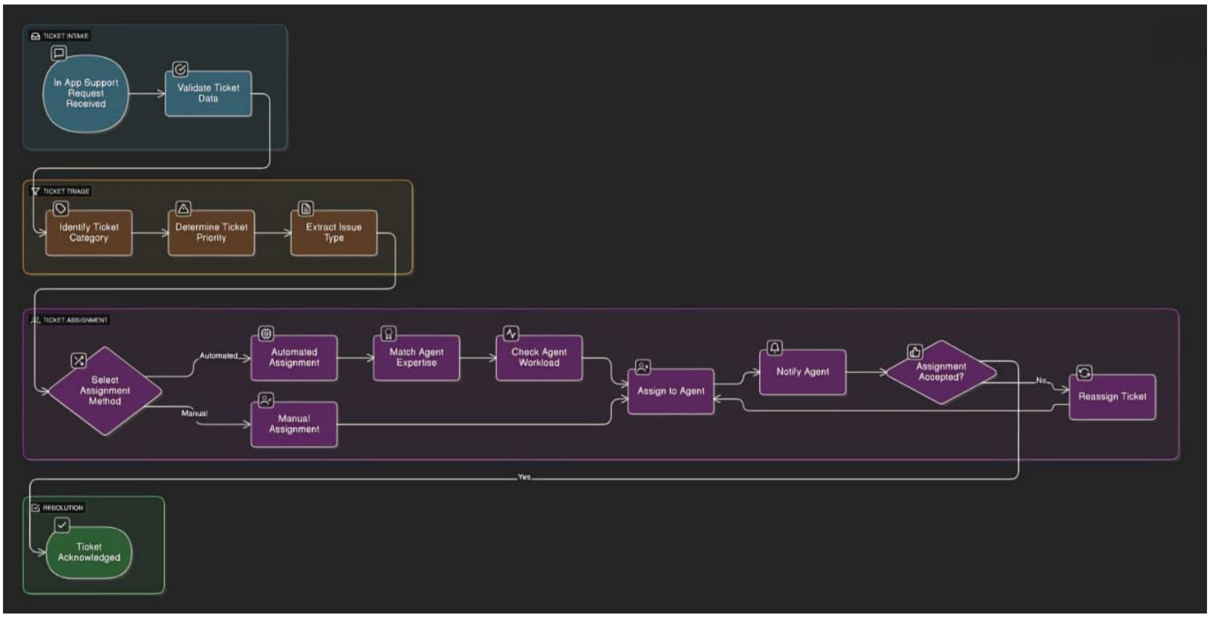
11. Deliverables of Design Phase

Deliverable	Description
System Architecture Diagram	Shows system components and integrations
Process Flowcharts	Maps ticket journey from creation to resolution
Data Model Diagram	Defines database structure and relationships
UI Wireframes	Visual layout of dashboards and control panels
AI Model Specification	Design of NLP classification and routing logic
Security Design Document	Defines access, encryption, and compliance requirements
Prototype Blueprint	Plan for initial working model



TICKET INTAKE
In App Support Request Received
Validate Ticket Data
TICKET TRIAGE
Identify Ticket Category
Determine Ticket Priority
Extract Issue Type
TICKET ASSIGNMENT
Select Assignment Method
Automated
Manual
Automated Assignment
Match Agent Expertise
Check Agent Workload
Manual Assignment
Assign to Agent
Notify Agent
Assignment Accepted?
No
Reassign Ticket
Yes
RESOLUTION
Ticket Acknowledged